



Northern University

Of Business and Technology Khulna

Project Proposal

Course Code: CSE 4204

Course Title: Mobile Computing Lab

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1. Project Title

Enterprise Nexus: Multi-Agent AI Business Automation OS with Security Intelligence

2. Problem Statement

Modern organizations face major challenges in managing recruitment, customer support, finance tracking and cybersecurity operations efficiently. Traditional ERP and CRM systems require significant manual effort, lack intelligent automation and fail to provide real-time decision-making capabilities. Existing solutions are often expensive, fragmented and unable to integrate AI-driven insights effectively.

Organizations generate large volumes of operational data every day, including resumes, customer tickets, invoices, emails and internal reports. Managing these processes manually increases operational cost, delays decision-making, and introduces security risks such as phishing attacks, spam submissions and financial fraud.

Existing systems generally focus on isolated functionalities and do not provide a unified AI-powered platform capable of intelligent automation, real-time analytics and centralized workflow orchestration.

Enterprise Nexus aims to solve these limitations by introducing a multi-agent AI architecture that automates enterprise workflows using intelligent AI agents, workflow orchestration, cybersecurity analysis and executive-level operational intelligence.

3. Objectives

The primary objectives of Enterprise Nexus are:

- To automate enterprise back-office operations using AI agents
- To develop an intelligent workflow orchestration system using n8n
- To implement AI-based phishing and malicious content detection
- To automate HR recruitment and candidate screening processes
- To develop a real-time customer sentiment analysis engine
- To implement AI-powered invoice processing and anomaly detection
- To create a real-time executive analytics dashboard
- To demonstrate practical enterprise-level AI integration

4. Proposed Features

Module 1: Security & Data Ingestion Filter - “The Ingestion Guard”

Features

- Phishing and spam analysis
- NLP-based spam confidence scoring
- Malicious script detection

- File quarantine system
- Live security alerts

Module 2: HR & Talent Acquisition Agent - “The HR Agent”

Features

- Autonomous CV parsing
- Skill and experience extraction
- Semantic job matching
- AI-based candidate compatibility scoring
- Automated interview scheduling emails

Module 3: Customer Support & Sentiment Engine - “The Support Agent”

Features

- Real-time sentiment analysis
- Emotion detection (Happy, Neutral, Furious)
- RAG-based support response generation
- Smart ticket prioritization
- Intelligent ticket routing

Module 4: Finance & Risk Auditing Agent - “The Finance Agent”

Features

- Multimodal invoice OCR
- Expense data extraction
- Duplicate invoice detection
- Financial anomaly analysis
- Automated admin alerts

Module 5: Executive Dashboard & Intelligence Hub - “The Executive Agent”

Features

- AI-generated executive summaries
- Natural language command center
- Real-time analytics dashboard
- Live workflow monitoring
- Interactive charts and insights

5. System Working Procedure

The system operates in three major phases:

Phase 1: Data Input & Security Filtration

1. Users upload files or text through the frontend or email system.
2. The Security Agent scans incoming data.
3. Malicious or suspicious files are isolated in the Quarantine Zone.
4. Safe data proceeds to workflow orchestration.

Phase 2: AI Orchestration & Collaboration

1. Clean data reaches the central n8n webhook.
2. n8n identifies the data type.
3. Tasks are routed to the appropriate AI agent.
4. Google Gemini API processes the requested task.
5. Structured results are stored in PostgreSQL.

Phase 3: Real-Time Dashboard Synchronization

1. Backend services trigger WebSocket events.
2. The frontend dashboard updates instantly.
3. Charts, analytics, and alerts refresh in real time.
4. Executive Agent generates operational summaries.

6. Technology Stack

Category	Technologies
Frontend	Next.js, React.js, Tailwind CSS
Backend	Node.js, Express.js
Database	PostgreSQL, Redis
AI Integration	Google Gemini API
Workflow Automation	n8n
Real-Time Communication	WebSockets
Data Visualization	Recharts, D3.js
Security Analysis	NLP Filtering
Authentication	JWT / OAuth
Deployment	Vercel, Render, Docker

7. AI Integration

Artificial Intelligence plays a central role in Enterprise Nexus.

The system uses AI for:

- Spam and phishing detection
- Resume parsing and candidate evaluation
- Sentiment analysis of customer tickets
- Retrieval-Augmented Generation (RAG) support responses
- Invoice OCR and financial anomaly detection
- Executive-level operational summaries

Google Gemini API will be used as the primary AI processing engine for text analysis, multimodal processing and intelligent response generation.

8. Expected Outcome

The proposed system is expected to:

- Reduce manual administrative workload
- Improve recruitment efficiency through AI-based candidate screening
- Enhance customer support prioritization
- Detect financial anomalies automatically
- Improve enterprise cybersecurity
- Provide real-time operational intelligence
- Demonstrate practical enterprise AI automation
- Improve workflow efficiency through centralized orchestration

9. Innovation & Uniqueness

Enterprise Nexus introduces several innovative capabilities:

- Multi-Agent AI collaboration
- Centralized workflow orchestration
- AI-powered enterprise automation
- Intelligent cybersecurity filtering
- Real-time executive analytics
- Natural language business query system
- Integrated HR, finance, support, and security ecosystem

Unlike traditional ERP systems, Enterprise Nexus combines AI intelligence, automation, cybersecurity and real-time analytics into a unified platform.

10. Initial Project Planning

Development Phase Tasks

Phase 1	Requirement analysis and system design
Phase 2	Frontend and backend setup

Development Phase Tasks

Phase 3	Database architecture development
Phase 4	AI integration using Gemini API
Phase 5	n8n workflow orchestration
Phase 6	Dashboard and analytics development
Phase 7	Testing and debugging
Phase 8	Final deployment and presentation

11. GitHub Repository Plan [GitHub Link: [Github](#)]

The GitHub repository will include:

- Frontend source code
- Backend APIs
- Database schema
- AI integration modules
- n8n workflow files
- README documentation
- Team collaboration setup

12. Conclusion

Enterprise Nexus represents a modern AI-powered enterprise automation platform capable of integrating intelligent workflow orchestration, cybersecurity analysis, customer support automation, recruitment intelligence and financial auditing into a single unified ecosystem.

The project demonstrates the practical application of artificial intelligence in enterprise software development and provides a scalable prototype for future AI-assisted business management systems.

This project fulfills the objectives of the Mobile Computing Lab course by combining AI integration, real-time computing, workflow automation, and modern software engineering practices into a professional enterprise solution.